ACADEMIC RESEARCH PROJECT (MBA-WILP)

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE PILANI

EC1 Experiential Learning – **Situated Learning Project**

AI Application Ecosystem (S1-25_MBACCZG580)

PHASE 3

Comprehensive AI Solution Blueprint, Governance, Lifecycle Plan & Roadmap

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Executive Summary (Phase-3)

This Phase-3 report presents a complete managerial blueprint for implementing an AI-enabled orchestration layer that improves the RFQ (Request for Quotation) response process used in automotive engineering services. Earlier phases highlighted how RFQs contain large, unstructured, bilingual documents, complex engineering dependencies, and require coordination across multiple global teams. These factors make the process slow, error-prone, and highly dependent on manual effort.

The proposed AI solution combines two core pathways inside a unified workspace:

(1) Document Intelligence with SME Routing and

(2) AI-Assisted Proposal Drafting.

Together, they automate RFQ intake, extract requirements, identify missing sections, detect anomalies, propose SME assignments and timelines, and generate first-draft text for human approval. This supports engineers and managers without replacing them, ensuring all final decisions remain human-controlled.

This Phase-3 submission builds on the understanding developed in Phase-1 and Phase-2. It incorporates engineering-specific complexities, cross-geo collaboration realities, and a clear decision-ownership model. The report outlines how the solution will be designed, validated, deployed, governed, and scaled over a 12–18 month period. Expected benefits include faster turnaround times, fewer rework loops, stronger accountability, and higher proposal quality for global automotive OEM customers.

The overall objective is to provide a practical, ethical, secure, and scalable AI blueprint aligned with organizational readiness and the Experiential Learning (EC1) requirements.

Context from Phase-1 and Phase-2

Phase-1 identified coordination gaps in the RFQ process—manual parsing, inconsistent SME routing, cross-region delays, and engineering dependencies across MIL, SIL, HIL, and VIL.

Phase-2 expanded this into an AI-enabled use-case with dual pathways (Document Intelligence + Drafting) to improve extraction, routing, and proposal consistency.

Phase-3 builds on these insights and presents a full implementation blueprint, governance model, and 12–18 month roadmap.

Acknowledgement

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Abbreviation Table

Abbreviation	Full Form
RFQ	Request for Quotation
OEM	Original Equipment Manufacturer
CFT	Cross-Functional Team
SME	Subject Matter Expert
CM%	Contribution Margin Percentage
TTL	Tata Technologies Limited
AI	Artificial Intelligence
RASIC	Responsible, Accountable, Support, Informed, Consulted
HW-SW	Hardware-Software
JSON	JavaScript Object Notation
CSV	Comma-Separated Values
PDF	Portable Document Format
KPI	Key Performance Indicator
EM	Engagement Manager

Title: AI-Enabled RFQ Orchestration for XiL (MIL/SIL/HIL/VIL) with Dual-Option Assistance.

1. End-to-End Lifecycle Blueprint

1.1 Problem Scoping & Requirement Definition

- The RFQ process for a German automotive OEM is complex because technical documentation is large, bilingual (German/English), multi-disciplinary, and tied to engineering dependencies such as MIL, SIL, HIL, and VIL stages.
- These dependencies align with global automotive engineering practices where large volumes of component, validation, and compliance data accumulate across long development cycles.
- Teams across India, Germany, and the UK work with different time zones and calendars, creating coordination friction and delays.
- In addition, automotive RFQs often include safety, cybersecurity, and functional compliance expectations similar to broader industry AI and process automation trends.
- The requirement for this solution is to streamline RFQ handling through AI-assisted extraction, routing, drafting, and decision tracking while keeping all approvals with human experts.

1.2 Data Preparation & Readiness Activities

- Data used in the RFQ process includes technical documents, Excel cost sheets, skill records, bilingual notes, and past proposal templates.
- Document intelligence techniques are widely used in automotive RFQ and RFP processing to extract structured information from large, inconsistent inputs.
- Before using AI, redacted proposal samples must be organised, section tags defined, SME availability mapped, and common compliance sections standardised.
- This aligns with best practices for AI workflow orchestration, where reusable components and structured data inputs improve consistency.

1.3 High-Level Model Logic (Conceptual Only)

- Document intelligence will extract key requirements, identify missing sections, detect bilingual content, and highlight large tables that need human confirmation. Routing logic determines who should work on specific sections by matching skills, workload, and time zones.
- Drafting logic creates first-draft text based on approved templates. Anomaly logic scans cost sheets to identify errors. Similar AI-supported automation workflows are used in procurement RFQs and supplier selection processes where unstructured RFQ content is converted into structured insights.

1.4 Validation, Business Testing & Performance Monitoring

- Validation occurs through four checkpoints: extraction review, routing approval, drafting review, and cost anomaly review.
- AI orchestration best practices emphasise human review, error-handling, and conditional logic as part of enterprise-grade workflow oversight.
- Performance monitoring uses dashboards showing delays, SME load balance, dependency risks, and draft completeness.
- To address the Phase-2 feedback, a single decision-ownership model is used: The **Engagement Manager (EM)** is the final authority when routing logic and drafting logic conflict.

  • If routing suggests SME A but drafting requires SME B first → EM decides sequence.
  • If timelines from two AI pathways clash → EM resets SLA using dashboard insights.
  This provides the unified conflict-resolution model expected in time-critical RFQs.

1.5 Deployment into Workflow

- Deployment uses existing enterprise tools for intake automation and workflow routing.
- AI-orchestrated RFQ automation is now common in global manufacturing and improves speed and consistency while retaining human oversight.
- When an RFQ arrives, a new workspace is created automatically.
- Tasks are assigned, drafts are generated, and all suggestions require explicit acceptance. The workspace maintains version control and decision logs.

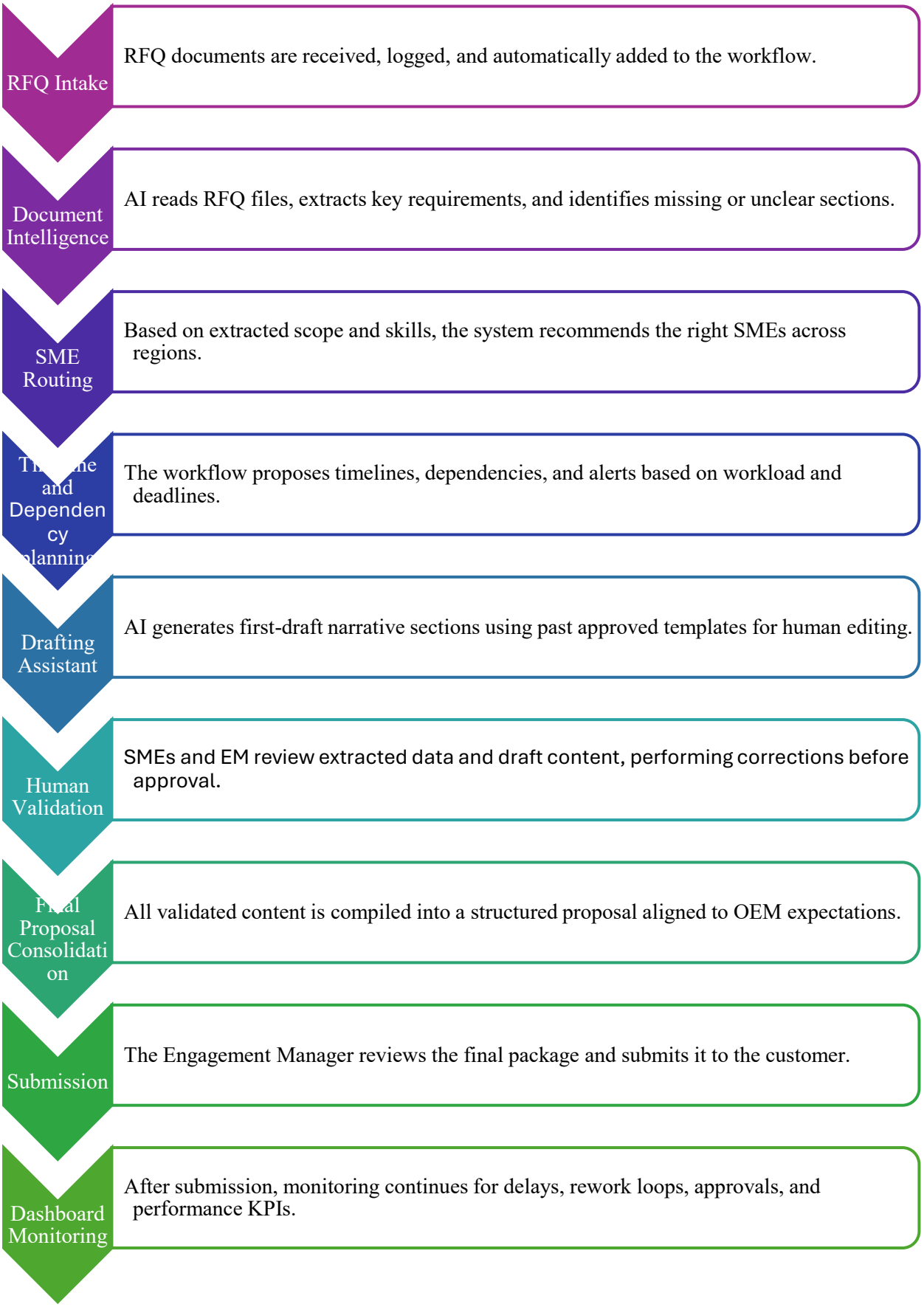
1.6 Post-Deployment Monitoring & Feedback Loops

- Continuous improvement follows a loop of: weekly review meetings, feedback from SMEs, error analysis, updates to templates, optimisation of routing rules, and periodic governance reviews.
- Cross-geo scheduling improvements can be inspired by intelligent scheduling approaches used in enterprise collaboration contexts.

Workflow

RFQ Intake → Document Intelligence → SME Routing → Timeline & Dependency Planning → Drafting Assistant → Human Validation → Final Proposal Consolidation → Submission → Dashboard Monitoring.

Diagram 1 – End-to-End AI-Assisted RFQ Workflow



2. Build vs Buy vs Partner Decision

2.1 Decision Criteria

- Key criteria include deployment time, cost, security, integration with enterprise environment, and long-term maintainability.
- AI-supported RFQ automation platforms in procurement and engineering typically weigh similar trade-offs between custom build, platform adoption, and partner-led implementation.

Comparison Table

Criteria	Build	Buy	Partner
Customization	Very High	Limited	High
Time to Deploy	Slow	Fast	Medium
Cost	High	Subscription	Service-based
Security	High	Mature enterprise controls	Medium
Best Use Case	Mature internal AI team	Pilot & early rollout	Complex integration

2.2 Recommended Strategy

- A hybrid approach:
- Start with “Buy” for fast pilot and secure enterprise-grade AI models.
- Involve a Partner to tailor templates and workflows.
- Transition selected components to “Build” once internal capability strengthens.

3. Governance & Responsible AI Practices

3.1 Human Oversight Mechanisms

- Consistent with best practices for AI orchestration, all AI recommendations require human review.
- The EM remains the central owner for decisions, sequencing, conflict resolution, and final approval.

3.2 Approvals, Review Boards & Compliance

- Two committees ensure oversight:
- AI Steering Committee for policy, data governance, and security.
- Proposal Review Council for technical, commercial, and compliance accuracy.

Fairness, Transparency & Accountability

- Routing explanations – "why this SME".
- Timeline reasons – "why this due date" are included in the decision log.
- Load-balance dashboards reduce bias toward specific SMEs.

3.4 Sensitive Data Protection

- Enterprise systems, encrypted storage, audit logs, and role-based access support secure processing.
- Data orchestration and governance approaches align with automotive AI best practices where large volumes of sensitive engineering data must be handled securely.

Governance RASIC Model

Activity	EM	SME	Finance	IT	Compliance	Delivery
Proposal Approval	A	C	C	I	I	R
Routing Overrides	R	C	I	I	I	C
Template Updates	C	R	I	C	I	A
NDA/GDPR Checks	I	I	I	C	R	A

4. Organizational Readiness & Capability Enablement

4.1 Teams Involved

- Engagement Managers, SMEs across MIL/SIL/HIL/VIL, Finance, IT, Compliance, and senior Delivery leadership.

4.2 Skills Required

- Users need skills in AI-assisted review, workflow navigation, technical validation, and understanding structured templates.
- This aligns with how AI systems assist engineering, sourcing, and procurement teams in modern manufacturing workflows.

4.3 Training & Enablement

- Includes micro-learning modules, role-based sessions, and scenario-based exercises using real OEM RFQs.

4.4 Building Trust for Non-Technical Users

- Trust is built through transparency, human approvals, clear dashboards, and documented decisions.

Skills & Responsibilities Matrix

Team	New Skills Required	Role in AI Workflow
Engagement Manager	AI approval, conflict resolution, dashboard use	Final decision owner
SMEs (MIL/SIL/HIL/VIL)	Extraction validation, draft editing	Technical reviewers
Finance	Anomaly checks	Cost validation
IT	Workflow support	Tool & access management
Compliance	Data checks	NDA & GDPR oversight

5. Change Management & Adoption Strategy

5.1 Communication Strategy

- Communication focuses on benefits such as reduced manual effort, faster proposal cycles, and better visibility.

5.2 Training, Pilot & Feedback Loops

- Pilot improvements follow short cycles, inspired by best practices where real-time user inputs shape AI-driven RFQ workflows.

5.3 Handling Resistance & Job-Impact Concerns

- Humans remain in charge. AI only assists. Clear messaging reduces fear and increases acceptance.

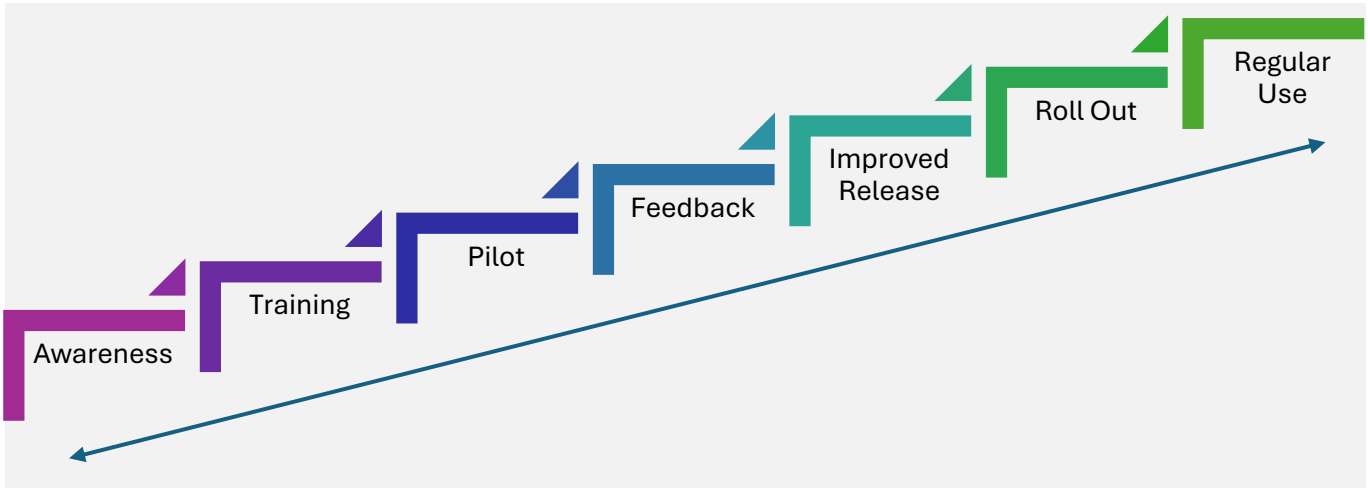
5.4 Embedding AI into Daily Work

- The workspace becomes the primary environment for RFQ tasks, replacing email-heavy processes.

Process flow

Awareness → Training → Pilot → Feedback → Improved Release → Rollout → Normal Use

Diagram 3 – Adoption Flow



Change Management and Adoption Flow illustrating awareness, training, pilots, feedback loops, and rollout stages

6. Scaling Strategy & Long-Term Evolution

6.1 Expansion Plan

- Begin with the German automotive OEM, then expand to other automotive OEMs and adjacent processes.
- Scaling aligns with broader trends in automotive AI adoption.

6.2 KPIs for Scaling

- 30–40% reduction in turnaround.
- 20% reduction in rework.
- 80% user satisfaction
- Zero critical compliance findings

6.3 Update / Replace / Sunset Triggers

- Triggered by template changes, regulation updates, or increased internal AI maturity.

Scaling KPI Matrix

KPI	Target	Source
Turnaround Time	30–40% improvement	Dashboard
Rework Loops	–20%	Logs
User Satisfaction	≥80%	Surveys
Compliance	Zero critical	Audits

7. 12–18 Month Roadmap

7.1 Phases & Milestones

- Months 1–2: Planning, data readiness
- Months 3–5: Pilot 1
- Months 6–8: Pilot 2
- Months 9–12: Rollout
- Months 13–18: Scaling to more OEMs

7.2 Timeline

- M1–M2: Planning
- M3–M5: Pilot 1
- M6–M8: Pilot 2
- M9–M12: Rollout
- M13–M18: Scaling

7.3 Resource Needs (People, Data, Tools)

- People: EM, SMEs, workflow developer, AI specialist, Finance, IT, Compliance.
- Data: RFQ content, SME registry, redacted proposals.
- Tools: Document intelligence, workflow automation, dashboards.
- Infrastructure: Secure enterprise environment with audit logging and role-based access

7.4 Risks & Mitigation

- Extraction errors → SME review.
- Routing bias → Load balancing.
- Time zone delays → Automated handover.
- Adoption issues → Continuous training

Risk and mitigation matrix

Risk Category	Description	Severity	Mitigation Strategy
Extraction Errors	Misreading engineering content, complex tables, or bilingual text.	High	Mandatory SME extraction review; iterative improvement of templates; highlight zones requiring manual checks.
Routing Bias	AI repeatedly assigns tasks to the same SMEs.	Medium	Load-balance dashboard; EM routing override; skill-weight limits.
Time-Zone Delays	India–Germany–UK workflows cause handover gaps.	Medium	Automated handover notes; overlapping meeting windows; escalation alerts.
Template Variations	OEM template changes cause confusion.	Medium	Template library updates; EM review checkpoints; standardized section tags.
User Adoption Resistance	SMEs hesitate to trust AI suggestions.	Low	Awareness sessions; small pilots; positive reinforcement from quick wins.
Data Governance Issues	Errors in storing sensitive RFQ files.	Low	Role-based access, audit logs, encryption, compliance review.

7.5 Expected Business Value

- Faster RFQ cycles, improved first-time-right quality, better accountability, stronger customer confidence, and reduced operational overhead.

End-to-End AI-enabled RFQ workflow showing how extraction, routing, drafting, and validation are orchestrated through a unified workspace

Roadmap Summary by Quarter

Phase	Duration	Key Activities	Outputs
Planning	M1–M2	Scope, approvals, data prep	Pilot charter
Pilot 1	M3–M5	RFQ automation tests	Initial results
Pilot 2	M6–M8	Multi-geo rollout	Improved logic
Rollout	M9–M12	Enterprise adoption	Governance model
Scaling	M13–M18	OEM expansion	Improved maturity

What Success Looks Like at 6 Months

- Pilot RFQs handled with 20–25% faster turnaround.
- Document intelligence stable for major sections (scope, assumptions, annexures).
- Initial SME routing accuracy ≥70%.
- Users adopt the workspace for daily tracking instead of email chains.
- First set of bilingual extraction rules validated.
- EMs confidently overriding routing/drafting conflicts with audit logs.

What Success Looks Like at 12 Months

- 30–40% reduction in overall RFQ processing time for typical engineering RFQs.
- Drafting assistant covers 60–70% of boilerplate content for proposals.
- Cross-geo India–Germany–UK workflows stabilised with predictable handovers.
- Clear governance model operational (Steering Committee + Review Council).
- SME workload balance significantly improved, reducing burnout and delays.
- System ready for expansion to additional OEMs or additional engineering functions.

Conclusion

This Phase-3 report presents a clear, practical, and scalable plan to implement an AI-enabled orchestration layer for automotive RFQs. The proposed solution strengthens the end-to-end workflow by combining intelligent extraction, routing, drafting assistance, and structured human oversight. It directly addresses real challenges faced during multi-disciplinary engineering RFQs, including cross-geo delays, complex dependencies, template variations, and manual gaps.

With strong governance, training, and responsible AI practices, the system ensures accuracy, transparency, and trust. The roadmap outlines a realistic 12–18 month rollout path that begins with controlled pilots and gradually expands across regions and OEM programs. Once fully deployed, the solution is expected to deliver measurable improvements in turnaround time, first-time-right quality, workload distribution, and overall proposal consistency.

The blueprint reflects a balanced approach that combines automation with human expertise. It aligns AI adoption with organizational readiness and supports the larger strategic goal of building a more efficient, competitive, and future-ready engineering services workflow.

Responsible AI Control Points

Control Point	Description	Purpose
1. Extraction Review	SMEs validate extracted requirements, bilingual content, tables and assumptions.	Ensures technical accuracy and avoids misinterpretation.
2. Routing Approval	EM reviews AI-suggested SME routing, workload, and time-zone feasibility.	Maintains fairness, avoids overload, and ensures correct responsibility mapping.
3. Draft Validation	SMEs and EM review AI-generated draft text for tone, correctness, and OEM template alignment.	Ensures proposal quality and compliance with OEM expectations.
4. Final Proposal Sign-Off	EM approves the final consolidated proposal before submission.	Human-in-loop assurance and accountability for final delivery.

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8. Academic Declarations

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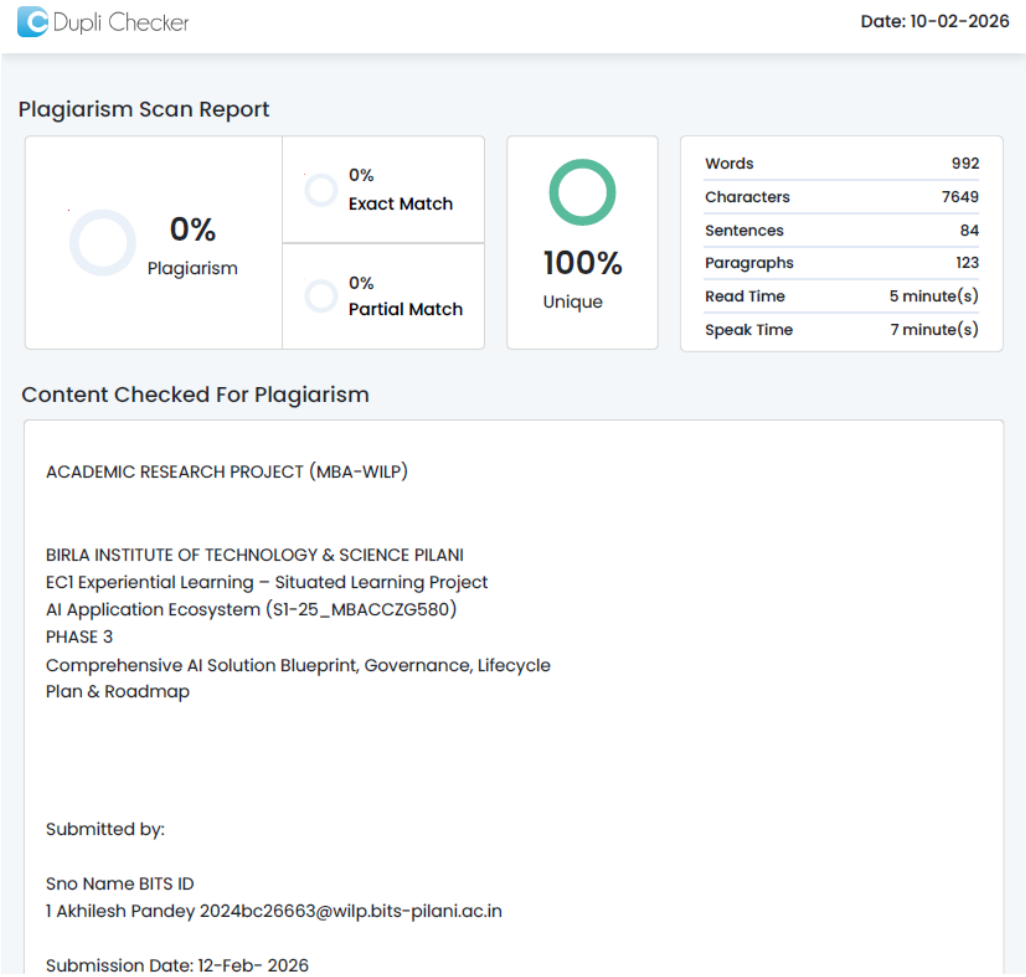

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